

33. Answer: D

For an ideal transformer,

$$P_{input} = P_{output}$$

$$I_{input} V_{input} = I_{output} V_{output}$$

$$(250 \text{ mA}) V_{input} = (250 \text{ mA})(9.0 \text{ V})$$

$$\therefore V_{input} = 27 \text{ V}$$

$$\frac{N_{output}}{N_{input}} = \frac{V_{output}}{V_{input}}$$

$$\therefore N_{input} = \frac{600 \times 27}{9.0} = 1800$$